



Fast-convergent Iterative Learning Control

Zeyu Li and Ying Tan

The Department of Mechanical Engineering

The University of Melbourne

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Outline

- Data-Driven Optimisation-Based ILC
- Fast Data-Driven Optimisation-Based ILC Schemes
- Summary

Data-Driven Optimisation-Based ILC



- Reformulates the tracking task as an optimization problem¹
- Different from the many optimisation-based ILC, in which models are known², we assume that the model is unknown
- Provides a framework for convergence-rate analysis

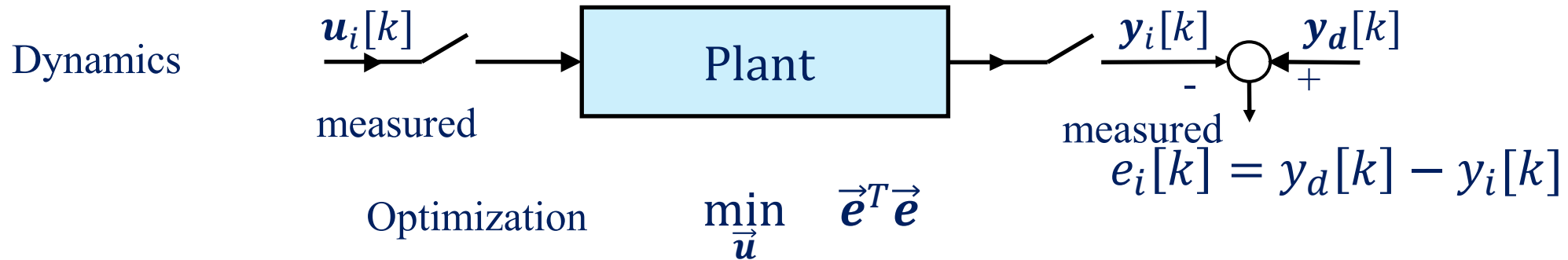
¹D. H. Owens, Iterative Learning Control: An Optimization Paradigm. London, U.K.: Springer-Verlag, 2016.

²C. T. Freeman and Y. Tan, "Iterative Learning Control With Mixed Constraints for Point-to-Point Tracking," IEEE Transactions on Control Systems Technology, 2013

Data-driven Optimisation-Based ILC

- ✓ Problem formulation: with a sample-data structure, an ILC design can be treated as an optimisation problem¹

$$\begin{aligned} x[k+1] &= f(x[k], u[k]) \\ y[k] &= g(x[k], u[k]) \end{aligned}$$



$$\begin{aligned} \vec{y}_d &= [y_d[0] \quad y_d[1] \quad \cdots \quad y_d[N-1]]^T & \vec{y}_i &= [y_i[0] \quad y_i[1] \quad \cdots \quad y_i[N-1]]^T \\ \vec{u}_i &= [u_i[0] \quad u_i[1] \quad \cdots \quad u_i[N-1]]^T & \vec{e}_i &= [e_i[0] \quad e_i[1] \quad \cdots \quad e_i[N-1]]^T \end{aligned}$$

¹D. H. Owens, Iterative Learning Control: An Optimization Paradigm. London, U.K.: Springer-Verlag, 2016.



Data-Driven Optimization-Based ILC

- If the dynamic models are known: $(f(\cdot, \cdot), g(\cdot, \cdot))$ are known, this becomes an optimisation problem

$$\begin{aligned}x[k + 1] &= f(x[k], u[k]) \\y[k] &= g(x[k], u[k])\end{aligned}$$

$$J(\vec{u}_i) = \vec{e}_i^T \vec{e}_i = \Phi_0(\vec{u}_i, x[0]) = \Phi(\vec{u}_i)$$



$$\min_{\vec{u}} \vec{e}^T \vec{e}$$

If $J(\cdot)$ is known, there are many optimisation algorithms off-the-shelf¹

$$\vec{u}_{i+1} = \vec{u}_i - \gamma_1 \frac{\partial J}{\partial \vec{u}}(\vec{u}_i)$$

$$\vec{u}_{i+1} = \vec{u}_i - \gamma_2 \frac{\partial J}{\partial \vec{u}}(\vec{u}_i) \left(\frac{\partial^2 J}{\partial \vec{u}^2}(\vec{u}_i) \right)^{-1}$$

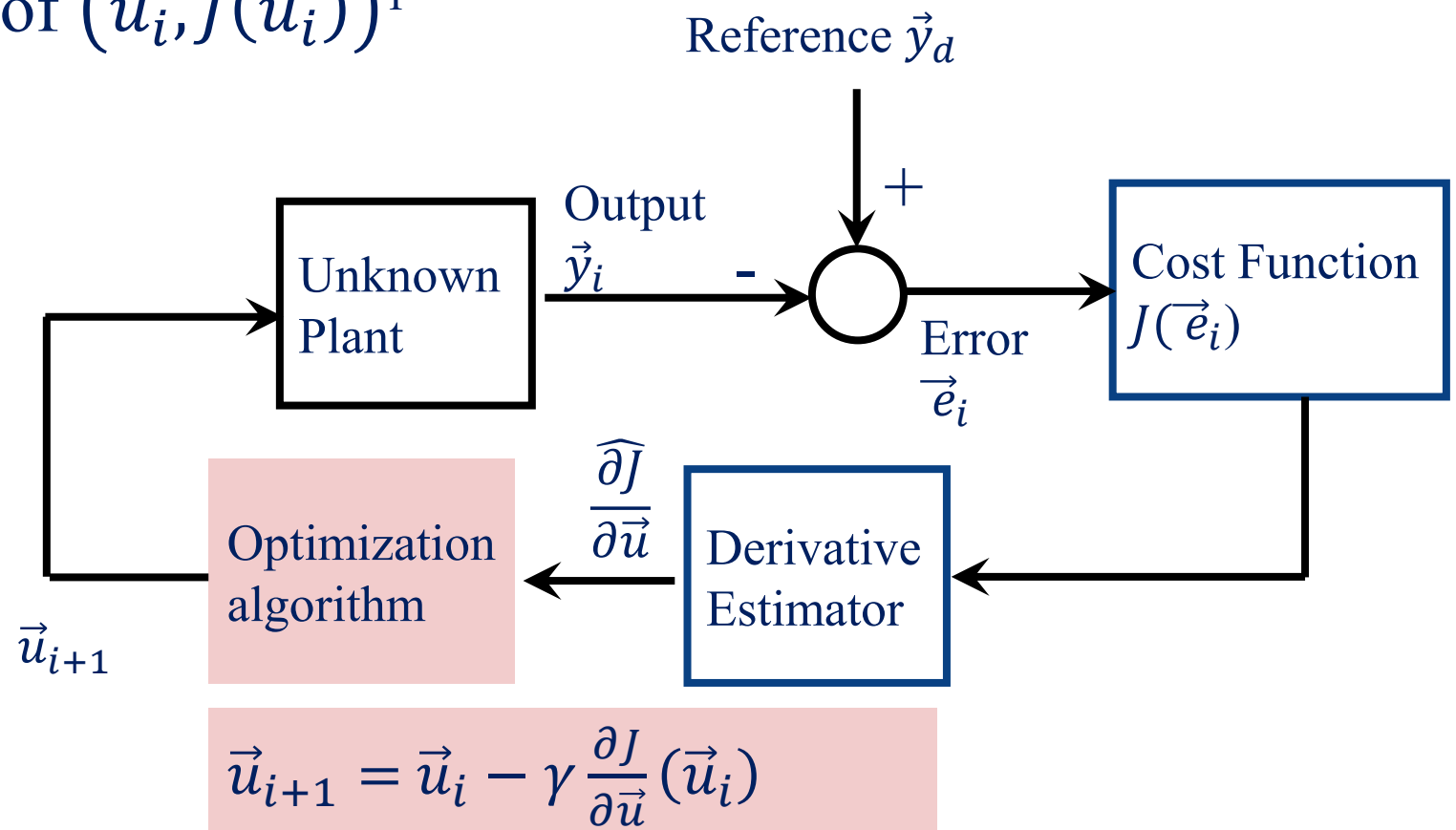
¹C. T. Freeman and Y. Tan, “Iterative Learning Control With Mixed Constraints for Point-to-Point Tracking,” IEEE Transactions on Control Systems Technology, 2013

Data-Driven Optimisation-Based ILC

- If we do not know the model, we can estimate the needed “derivatives” from the measurements of $(\vec{u}_i, J(\vec{u}_i))^1$

$$\vec{u}_{i+1} = \vec{u}_i - \gamma \frac{\partial J}{\partial \vec{u}}(\vec{u}_i)$$

$$\vec{u}_{i+1} = \vec{u}_i - \gamma \frac{\partial \widehat{J}}{\partial \vec{u}}(\vec{u}_i)$$



¹D. Nešić, Y. Tan, W. H. Moase, and C. Manzie, “A Unifying Approach to Extremum Seeking: Adaptive Schemes Based on Estimation of Derivatives”, 49th IEEE Conference on Decision and Control (CDC), 2010,

Data-Driven Optimisation-Based ILC

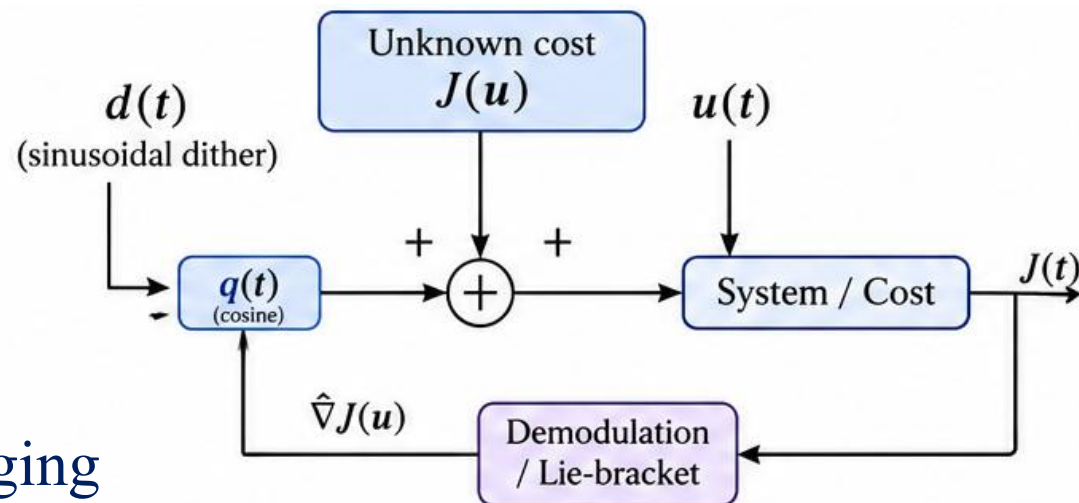
➤ Dither-based gradient estimation: using Lie-bracket¹

A simple example: $u \in R$ $\omega \gg 1$

Consider two periodic vector fields:

$$\dot{u} = \sqrt{\omega} [F_1(J(u)) \cos(\omega t) + F_2(J(u)) \sin(\omega t)]$$

Dithers cause $u(t)$ to oscillate rapidly. Averaging over the fast oscillations gives a slow averaged system that approximates the evolution of $u(t)$



Dither-based gradient estimation via Lie-bracket approximation

Averaged dynamics (Lie bracket):

$$\bar{u} \approx \frac{1}{2} [F_1, F_2](u)$$

(after averaging over a fast period)

¹H.-B. Dürr, M. S. Stanković, C. Ebenbauer, and K. H. Johansson, “Lie Bracket Approximation of Extremum Seeking Systems,” *Automatica*, 2013

Data-Driven Optimisation-Based ILC

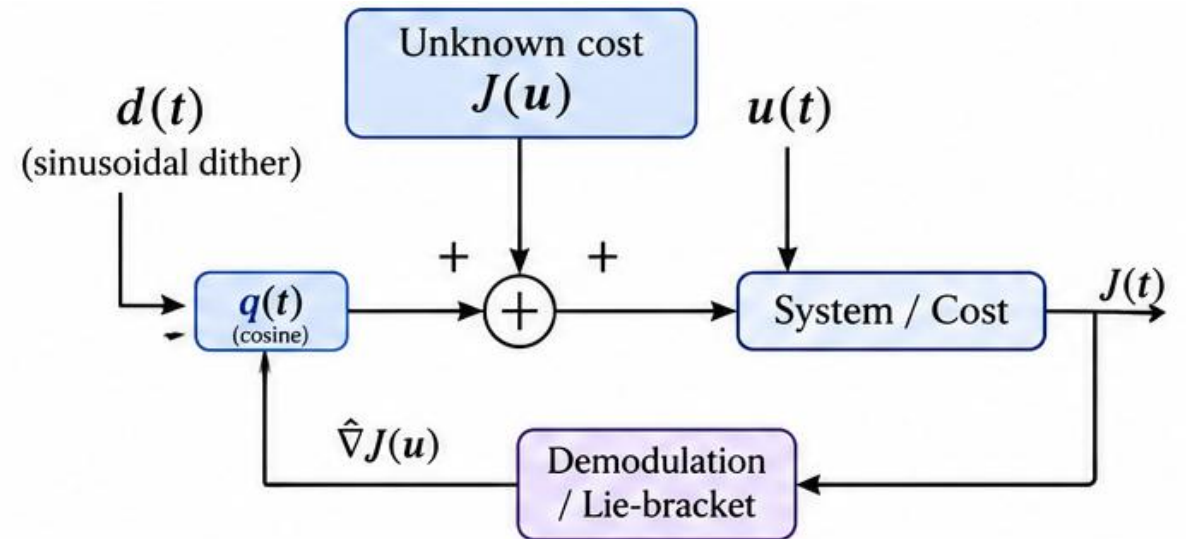
➤ Dither-based gradient estimation: using Lie-bracket

A simple example: $u \in \mathbb{R}$

Choose $F_1(J) = \cos J$, $F_2(J) = \sin J$

$$\Rightarrow \frac{1}{2} [F_1, F_2](u) = \frac{1}{2} \frac{dJ}{du}$$

$$\dot{u} \approx \frac{1}{2} \frac{\partial J}{\partial \vec{u}}(u)$$



Dither-based gradient estimation via Lie-bracket approximation

Fast dithers ➡ Averaging / Lie bracket ➡ Gradient direction

Data-Driven Optimisation-Based ILC

- If the control input is a high-dimensional input, $u \in R^m$, we have the following updating law

$$\vec{u}_{i+1}^a = \vec{u}_i^a + \sqrt{\Gamma}\varepsilon\cos(J(\vec{u}_i^a))\mathbf{d}_i + \sqrt{\Gamma}\varepsilon\sin(J(\vec{u}_i^a))\mathbf{q}_i,$$

where Γ is a diagonal matrix,

$0 < \varepsilon \ll 1$ is a small design parameter ,

$\mathbf{d}_i, \mathbf{q}_i$ are properly designed dithers

$\mathbf{d}_i$ and $\mathbf{q}_i$ are periodic signals (with a common period T), averaging to be zero over one period
They do not affect each other , for example,
 $\int_0^T \sin(\omega t) \cos(\omega t) dt = 0.$

Data-Driven Optimisation-Based ILC

- An assumption is needed to select appropriate dithers

Assumption 1 (Properly designed dithers):

(Let $d_{i,l}(d_{i,j})$, $q_{i,l}(q_{i,j})$ be the $l^{th}(j^{th})$ element of $\mathbf{d}_i$, $\mathbf{q}_i$ respectively.)

- $d_{i,l}$, $q_{i,l}$ are bounded periodical signals with the same period $K_l \in \mathcal{N}$.

- Both $d_{i,l}$ and $q_{i,l}$ have zero mean.

- Let K be the least common multiple of K_l . There exist positive constants $c_{1,l}$ and $c_{2,l}$, such that

$$\sum_{i=0}^{K-1} (q_{i,j} \sum_{s=0}^i q_{s,l}) = \begin{cases} c_{1,l}, & \text{if } j = l \\ 0, & \text{otherwise} \end{cases}$$

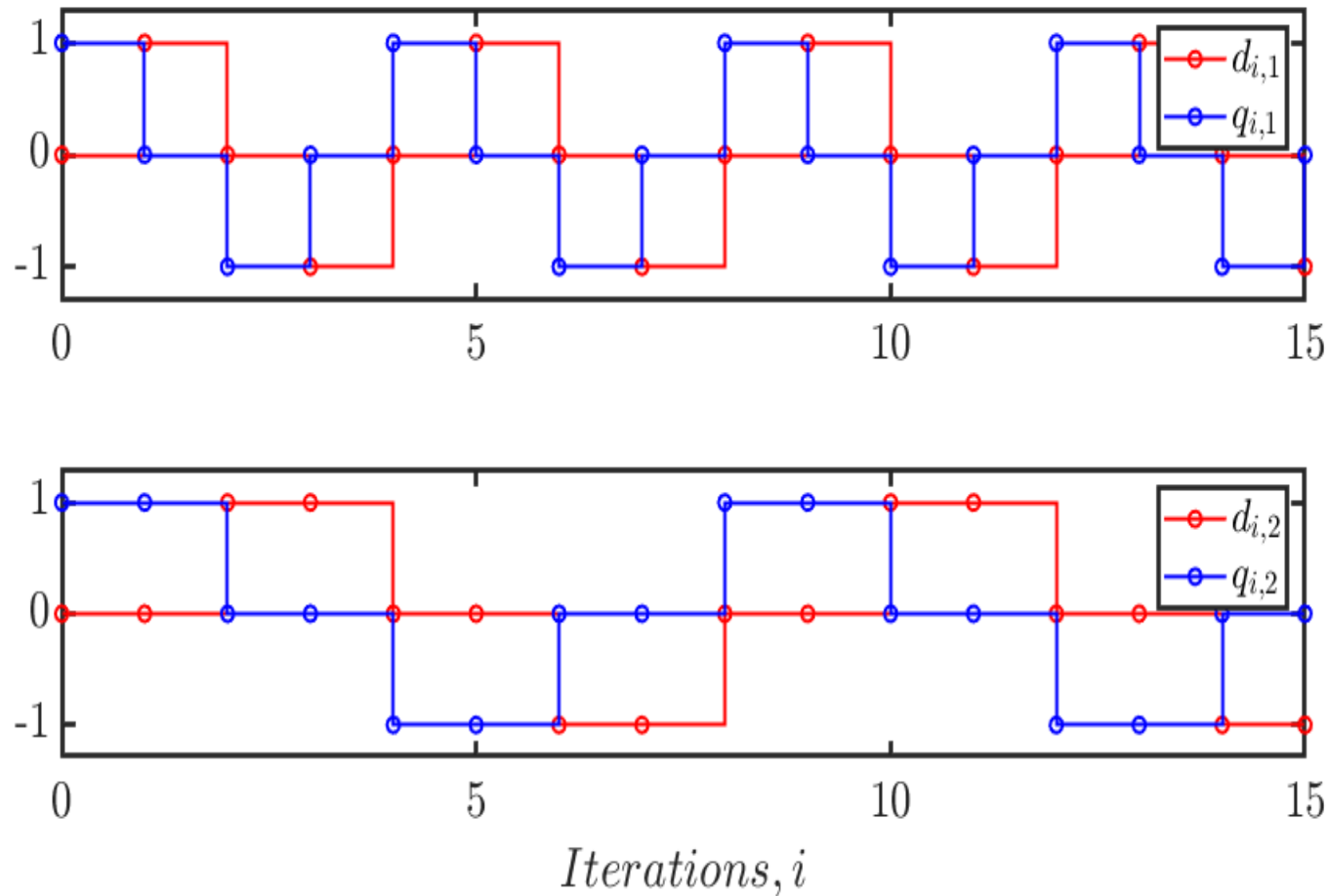
$$\sum_{i=0}^{K-1} (q_{i,j} \sum_{s=0}^i d_{s,l}) = \begin{cases} c_{2,l}, & \text{if } j = l \\ 0, & \text{otherwise} \end{cases}$$

$$\sum_{i=0}^{K-1} (q_{i,j} \sum_{s=0}^i d_{s,l}) = - \sum_{i=0}^{K-1} (d_{i,j} \sum_{s=0}^i q_{s,l}),$$

$$\sum_{i=0}^{K-1} (d_{i,j} \sum_{s=0}^i d_{s,l}) = \sum_{i=0}^{K-1} (q_{i,j} \sum_{s=0}^i q_{s,l}).$$

Data-Driven Optimisation-Based ILC

- An example of dithers



Data-Driven Optimisation-Based ILC

➤ Result 1¹:

Theorem 1: Under the dither conditions in Assumption 1, for sufficiently small $\varepsilon > 0$, the dither-based update,

$$\vec{\mathbf{u}}_{i+1}^a = \vec{\mathbf{u}}_i^a + \sqrt{\Gamma\varepsilon}\cos(J(\vec{\mathbf{u}}_i^a))\mathbf{d}_i + \sqrt{\Gamma\varepsilon}\sin(J(\vec{\mathbf{u}}_i^a))\mathbf{q}_i,$$

are close to the following gradient-descent algorithm,

$$\vec{\mathbf{u}}_{i+1} = \vec{\mathbf{u}}_i - \varepsilon \frac{\partial J}{\partial \vec{\mathbf{u}}}(\vec{\mathbf{u}}_i),$$

such that $|\vec{\mathbf{u}}_{i+1}^a - \vec{\mathbf{u}}_{i+1}| = O(\sqrt{\varepsilon})$.

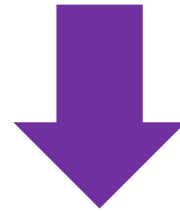
The dithers make the model-free algorithm behave like gradient descent

¹Li, Z., Sebastian, G., Tan, Y., & Oetomo, D. "On Iterative Learning Control for Nonlinear Discrete-Time Systems with Input and Output Constraints", In 2022 15th Asian Control Conference (ASCC), 2022.

Data-Driven Optimisation-Based ILC

➤ Remarks

- ✓ Properly designed dithers $\rightarrow$ Gradient estimate $\widehat{\frac{\partial J}{\partial \vec{u}}}(\vec{u}_i) \rightarrow$
Model-free gradient descent
- ✓ Smaller $\varepsilon \rightarrow$ Better gradient-descent approximation $\rightarrow$ Slower convergence
- ✓ It can also handle input and output constraints by introducing barrier functions



Data-driven Optimisation-Based ILC

Data-Driven Optimisation-Based ILC

- Challenges: Dither-based gradient estimation in high dimensions

$$\widehat{\frac{\partial J}{\partial \vec{u}}}(\vec{u}_i) \in R^{m \times N}$$

- ✓ Each dimension needs two dithers with one frequency
- ✓ We have $m \times N$ frequencies with $2m \times N$ dithers
- ✓ In the iteration domain, the dither frequency is limited. As the input dimension increases, more distinct dither frequencies are required. To maintain the scale separation required by averaging, the learning rate must be reduced, resulting in slower convergence (thousands to tens of thousands of iterations)

Data-Driven Optimisation-Based ILC

➤ Simulation Results

A nonlinear scalar discrete system is considered:

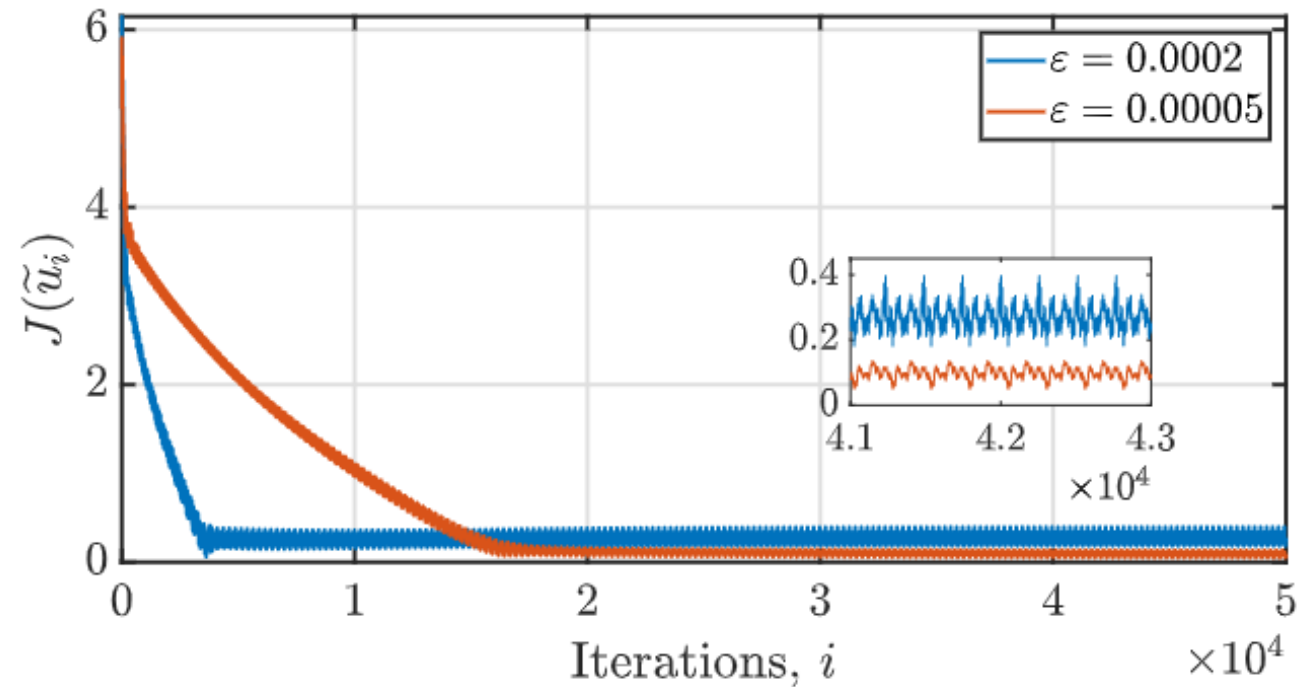
$$x[k + 1] = \sin(x[k]) + u[k]$$

$$y[k] = x[k] + u[k]$$

The reference input is $u_r[k] = \sin(2k)$, $\forall k \in [0, 1, \dots, 6]$.

The initial condition $x[0] = 0.8$.

$$m \times N = 7$$



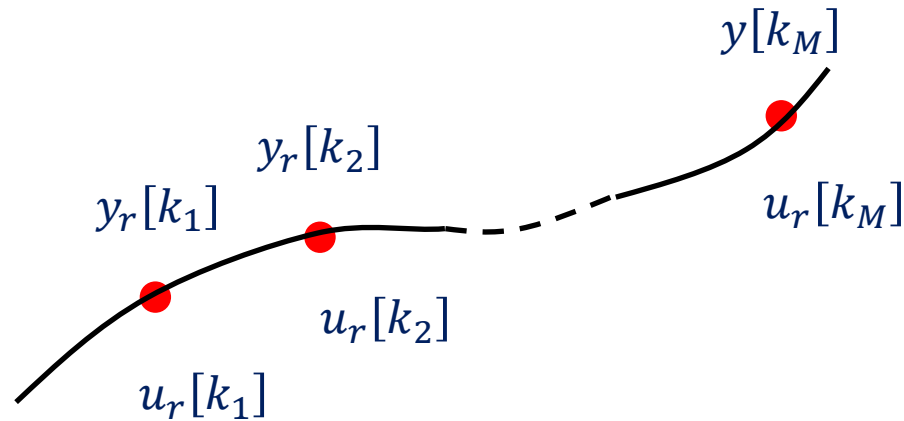


Fast Data-Driven Optimisation-Based ILC Schemes

- We want to improve the convergence speed for data-driven optimisation-based ILC
- Reduce the dimension of input $\vec{u}_i^a(d_i, q_i)$  Point-to-point ILC
- Fast Gradient descent algorithm

Fast Data-Driven Optimisation-Based ILC Schemes

- Now our focus is point-to-point (PtP) problem: (Only a few critical points are of interest)



New Cost
defined

$$J[k_l] = \mathbf{e}[k_l]^T \mathbf{e}[k_l]$$

$$\min_{\vec{u}} \sum_l^M J[k_l]$$

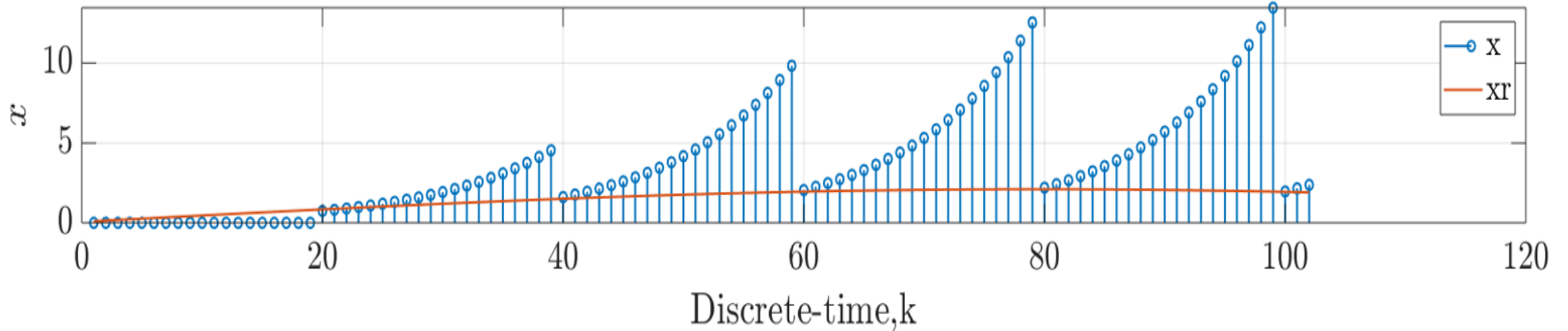
$$\vec{u}_r^s = [u_r[k_1] \quad u_r[k_2] \quad \cdots \quad u_r[k_M]]^T$$

Dimension of output tracking $N \rightarrow M$ (from N sampling points to M sampling points, $M \ll N$)

Fast Data-Driven Optimisation-Based ILC Schemes

- Can we update the control input only at the M target points?

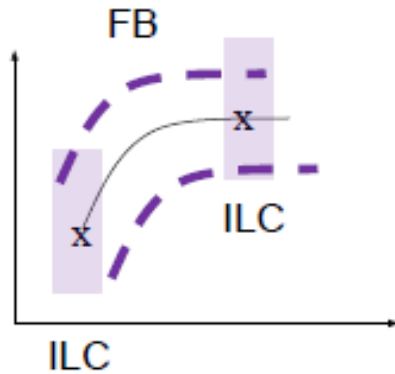
Tracking errors may grow between consecutive target points and propagate unboundedly



Fast Data-Driven Optimisation-Based ILC Schemes

- Feedback control is introduced to bound the tracking error between consecutive target points¹

Control strategy: $\vec{u}_i[k] = \vec{u}_i^{fb}[k]$ (Feedback) + $\vec{u}_i^{ff}[k]$ (Feedforward)



$$\vec{u}_i^{ff}[k] = \begin{cases} 0 & k \neq s \\ \vec{u}_i^s[k] & k = s, s = 1, \dots, M \end{cases}$$

Feedback control: Bounds the tracking error over the entire time interval, ensuring uniformly bounded errors at the target points during ILC learning

Data-driven ILC: Achieves high-precision tracking at the target points

¹ Li, Zeyu, PhD thesis. Optimization-Based Iterative Learning Control Framework with Practical Considerations, 2025



Fast Data-Driven Optimisation-Based ILC Schemes

- Feedback design requirement: The output-feedback controller must stabilize the system between consecutive target points.

Assumption 2: For the output feedback controller, there exists a Lyapunov candidate $V(\mathbf{e}): \mathcal{D} \subseteq \mathcal{R}^n \rightarrow \mathcal{R}$, such that the following inequalities are satisfied:

$$\alpha_1(|\mathbf{e}|) \leq V(\mathbf{e}) \leq \alpha_2(|\mathbf{e}|)$$

$$\frac{\partial V}{\partial \mathbf{e}_{\xi,0}} (\delta h(\mathbf{x}[k+1]) - D(\mathbf{x}[k+1])\mathbf{u}^{fb}[k+1]) - V(\mathbf{e}[k]) \leq -\alpha_3(|\mathbf{e}[k]|),$$

Where $\alpha_1(\cdot)$ - $\alpha_3(\cdot)$ are class $\mathcal{K}$ functions, $\frac{\partial V}{\partial \mathbf{e}_{\xi,0}} \triangleq \frac{\partial V}{\partial \mathbf{e}} \Big|_{\mathbf{e}=\mathbf{e}[k]+p(\mathbf{e}[k+1]-\mathbf{e}[k])}$ for some $p \in (0,1)$.



Fast Data-Driven Optimisation-Based ILC Schemes

➤ Convergence analysis¹

Theorem: If Assumption 1 and Assumption 2 hold, for a given pair (Δ, v) , such that there exist a positive constant ε_1^* , , such that for any $\varepsilon \in (0, \varepsilon_1^*]$, the feedforward control input at the time of interest satisfy

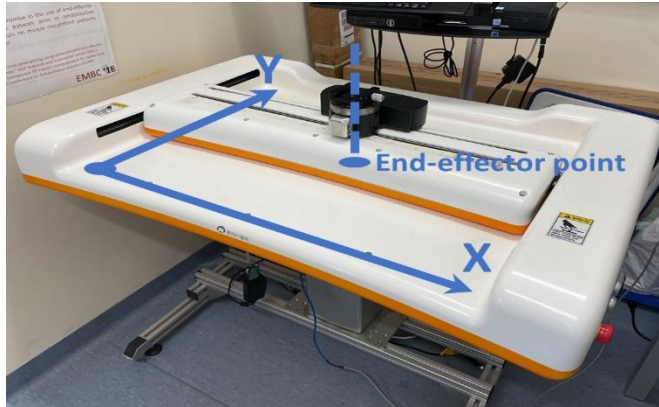
$$|\vec{\mathbf{u}}_i^s - \vec{\mathbf{u}}_r^s| \leq \max\{\beta(|\vec{\mathbf{u}}_0^s - \vec{\mathbf{u}}_r^s|, \varepsilon i), v\}$$

for all $i \in \mathcal{N}$ and $|\vec{\mathbf{u}}_0^s - \vec{\mathbf{u}}_r^s| \leq \Delta$.

¹ Li, Zeyu, PhD thesis. Optimization-Based Iterative Learning Control Framework with Practical Considerations, 2025

Fast Data-Driven Optimisation-Based ILC Schemes

➤ Experiment validation¹



The platform is an ArmMotus-M2 2-DOF planar robotic manipulator with two axes (X,Y), denoted as (y_1, y_2) . The workspace is $0.575\text{m} \times 0.401\text{m}$. The end-effector is driven by two brushless DC motors, and the measured outputs are the positions along the two axes.

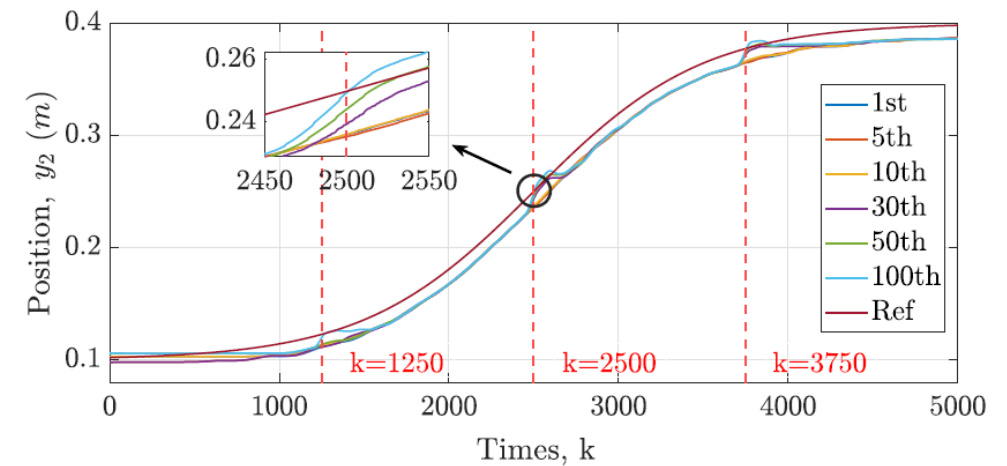
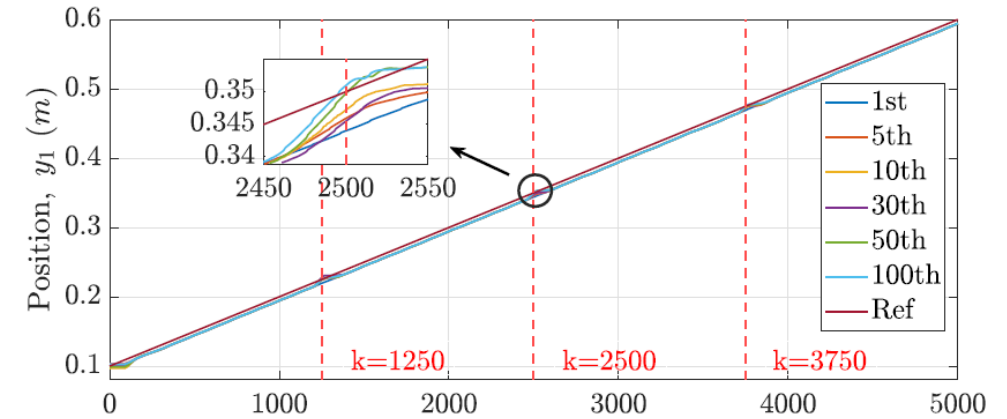
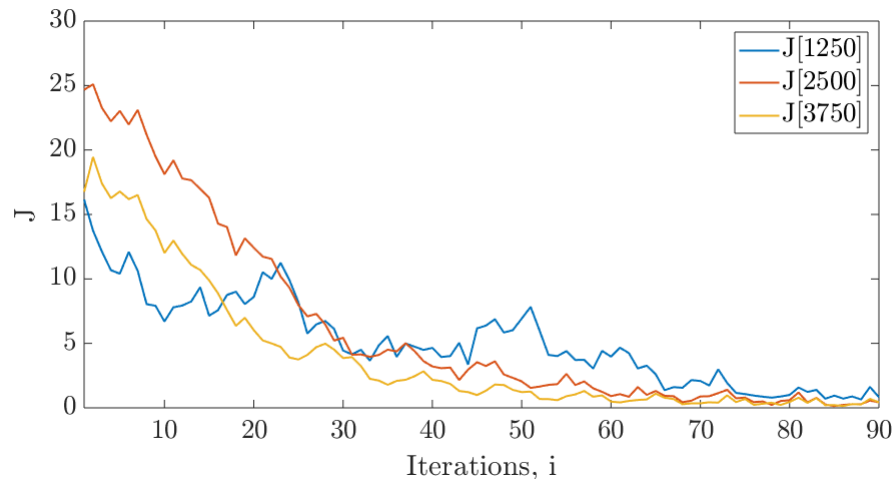
- ✓ 2-DOF planar robotic manipulator
- ✓ Trial duration: $T_f = 10\text{s}$
- ✓ Target times: 2.5, 5, 7.5s
- ✓ Sampling period: $\delta t = 0.002\text{s}$
- ✓ $\Delta = 0.02$, $v = 0.007$, and $\varepsilon = 0.01$ for this experiment

¹ Li, Zeyu, PhD thesis. Optimization-Based Iterative Learning Control Framework with Practical Considerations, 2025

Fast Data-Driven Optimisation-Based ILC Schemes

➤ Results

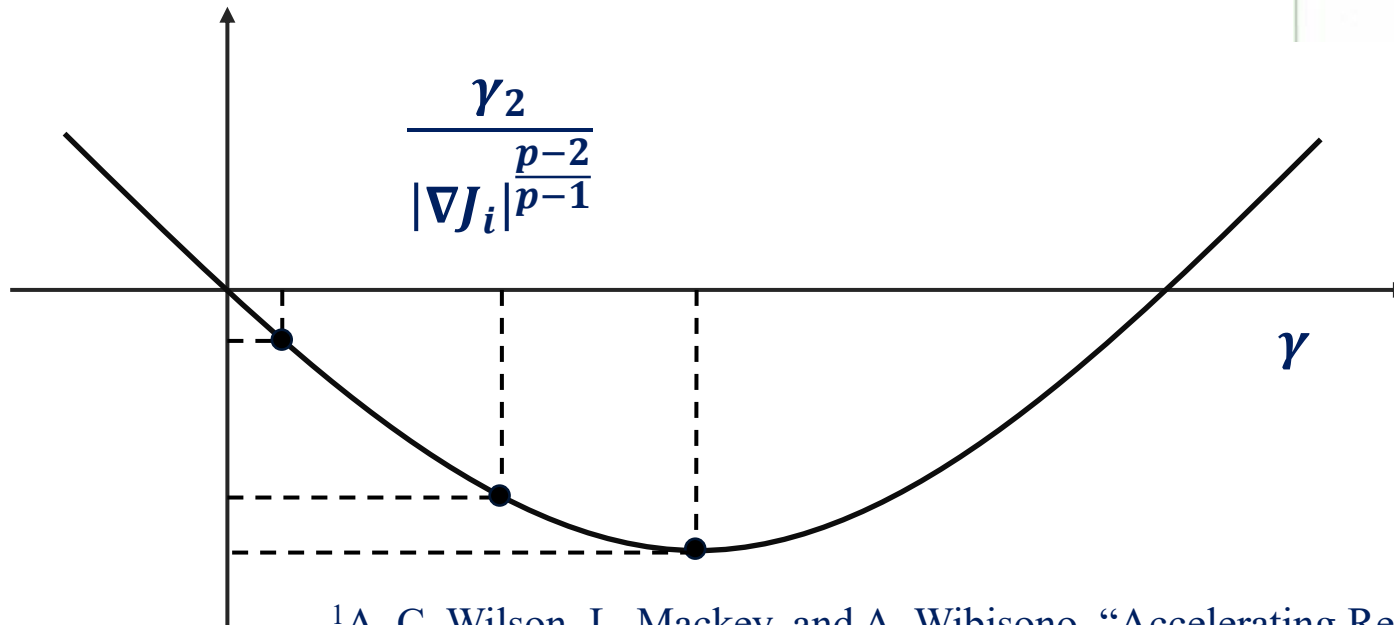
- ✓ Tracking errors remain bounded over the entire trajectory
- ✓ Tracking accuracy improves at the three target points
- ✓ The target-point costs converge with iteration (it converges within 90 iterations).



Fast Data-Driven Optimisation-Based ILC Schemes

➤ Fast gradient-based optimisation for ILC

RGD¹ accelerates gradient descent near the optimum, achieving a faster convergence rate $O\left(\frac{1}{i^{p-1}}\right), p \geq 3$, compared with $O\left(\frac{1}{i}\right)$ for standard gradient-descent



Rescaled Gradient Descent (RGD)

$$\vec{u}_{i+1} = \vec{u}_i - \frac{\varepsilon}{\|\nabla J_i\|^{\frac{p-2}{p-1}}} \nabla J_i, \quad p \geq 3$$

Adaptive (rescaled) step size: $\varepsilon'_i = \frac{\varepsilon}{\|\nabla J_i\|^{\frac{p-2}{p-1}}}$

Small gradient
 $\Rightarrow$ RGD rescales the gradient
 $\Rightarrow$ larger effective update
 $\Rightarrow$ faster convergence

¹A. C. Wilson, L. Mackey, and A. Wibisono, “Accelerating Rescaled Gradient Descent: Fast Optimization of Smooth Functions,” Advances in Neural Information Processing Systems (NeurIPS), vol. 32, 2019.

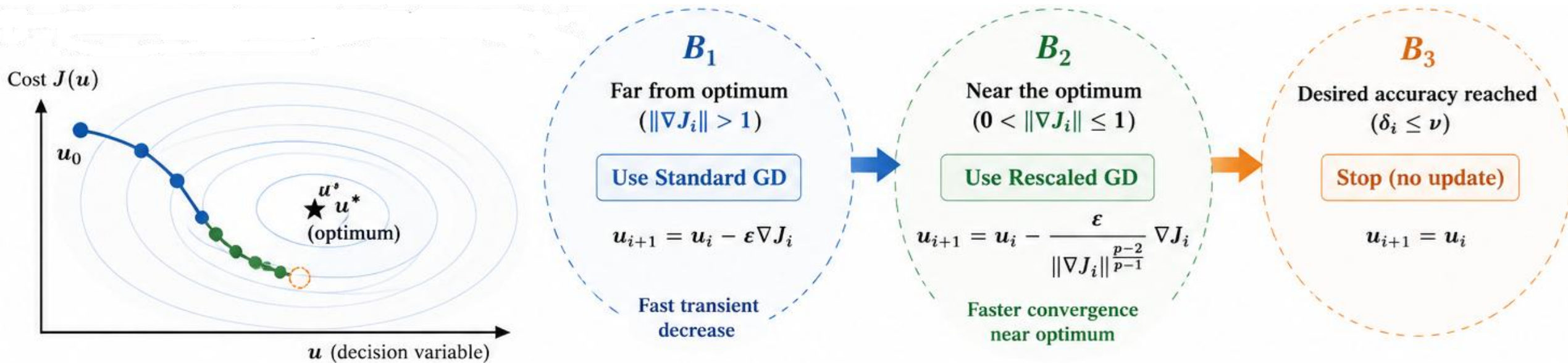
Fast Data-Driven Optimisation-Based ILC Schemes

- RGD is faster near the optimum, but may be slower during the transient

When is which faster?		
Gradient norm $g_i = \ \nabla J_i\ $	Compare update sizes	Faster method
$g_i > 1$	$\epsilon g_i > \epsilon g_i^{\frac{1}{p-1}}$ (since $\frac{1}{p-1} < 1$)	Standard GD (larger step)
$0 < g_i < 1$	$\epsilon g_i < \epsilon g_i^{\frac{1}{p-1}}$	RGD (larger step)
$g_i = 1$	$\epsilon g_i = \epsilon g_i^{\frac{1}{p-1}}$	Equal

Fast Data-Driven Optimisation-Based ILC Schemes

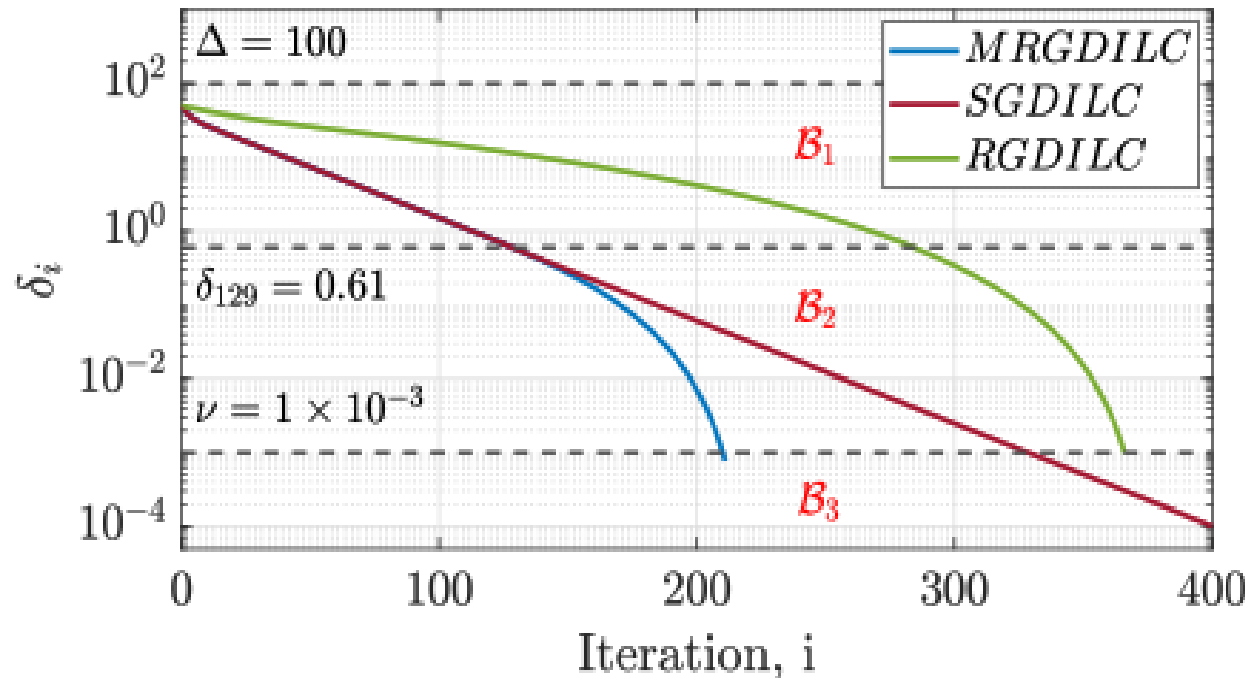
➤ Three-region strategy¹



¹ Li, Zeyu, PhD thesis. Optimization-Based Iterative Learning Control Framework with Practical Considerations, 2025

Fast Data-Driven Optimisation-Based ILC Schemes

➤ Simulation results¹



Modified RGD retains the fast transient behaviour of standard GD and accelerates convergence near the optimum (212 vs. 327 iterations).

$$\mathbf{x}[k+1] = \begin{bmatrix} 0.1 & 0 \\ 0 & 0.1 \end{bmatrix} \mathbf{x}[k] + \begin{bmatrix} 0.6 & 0 \\ 0 & 0.4 \end{bmatrix} \mathbf{u}[k],$$

$$\mathbf{y}[k] = \begin{bmatrix} 2 & 0 \\ 3 & 1 \end{bmatrix} \mathbf{x}[k] + \begin{bmatrix} 0 & 0 \\ 0 & 2 \end{bmatrix} \mathbf{u}[k].$$

$$y_{1,r}^d[k] = 4 \sin(0.4\pi k),$$

$$y_{2,r}^d[k] = 2 \sin(0.3\pi k).$$

¹ Li, Zeyu, PhD thesis. Optimization-Based Iterative Learning Control Framework with Practical Considerations, 2025

Summary

- **Data-driven optimisation-based ILC:** Dither signals estimate the gradient **without requiring a system model**
- **Faster learning:** P2P tracking reduces the optimisation dimension: $m \times N \rightarrow m \times M, M \ll N$ while feedback ensures bounded tracking
- **Faster optimisation:** Modified RGD accelerates convergence, particularly near the optimum



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Thank you

Background

Iterative Learning control (ILC) is widely used in the tracking problems of repetitive tasks. It has features:

- Finite time repetitive tasks
- High precision trajectory tracking



(Key Idea) Past tracking performance can be used to improve the tracking performance → perfect tracking on a finite time interval

ILC is widely used in pick and place^[1], chemical batch processing^[2], laser cutting^[3], ...

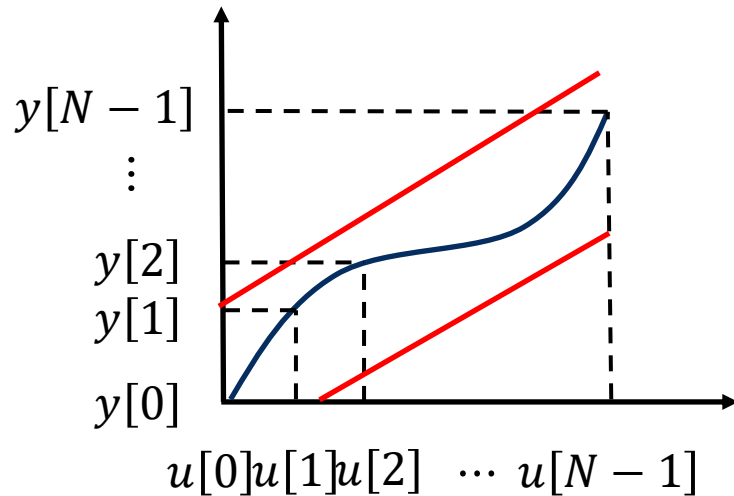
[1] Steinbuch, M. and van de Molengraft, R., 2000. Iterative learning control of industrial motion systems. IFAC Proceedings Volumes, 33(26), pp.899-904.

[2] Chen, Y., Xu, J.X., Lee, T.H. and Yamamoto, S., 1997, July. Comparative studies of iterative learning control schemes for a batch chemical process. In Proceedings of the IEEE Singapore Int. Symposium on Control Theory and Applications (pp. 166-170).

[3] Wallén, J., Norrlöf, M. and Gunnarsson, S., 2008. Arm-side evaluation of ILC applied to a six-degrees-of-freedom industrial robot. IFAC Proceedings Volumes, 41(2), pp.13450-13455.

Background (ILC Problem Formulation)

Constrained tracking problem with discrete-time system considered



The super vectors $\tilde{\mathbf{y}}, \tilde{\mathbf{u}}$ are defined as
 $\tilde{\mathbf{y}} = [\mathbf{y}[0], \mathbf{y}[1], \dots, \mathbf{y}[N-1]]^T \in \mathcal{R}^{m \cdot N}$
 $\tilde{\mathbf{u}} = [\mathbf{u}[0], \mathbf{u}[1], \dots, \mathbf{u}[N-1]]^T \in \mathcal{R}^{p \cdot N}$

Denote the tracking error $\tilde{\mathbf{e}} = \tilde{\mathbf{y}}_r - \tilde{\mathbf{y}}$

Control objective:

$$\min_{\tilde{\mathbf{u}}} \tilde{\mathbf{e}}^T \tilde{\mathbf{e}}$$

$$\text{s.t. } \begin{aligned} V_1(\mathbf{y}[k]) &\geq \mathbf{a}_1 \\ V_2(\mathbf{u}[k]) &\geq \mathbf{a}_2 \end{aligned}$$

Dynamics



Static mapping: $\tilde{\mathbf{y}} = Q(\tilde{\mathbf{u}}, \mathbf{x}[0])$



Optimization

Background (ILC Problem Formulation)

Constrained tracking problem with discrete-time system considered



Norm optimal iterative learning control (NOILC) ^{[3],[4]}: For $\tilde{\mathbf{y}} = Q(\tilde{\mathbf{u}}, \mathbf{x}[0])$,

if $Q(\cdot)$ is known $\tilde{\mathbf{u}}_{i+1} = \tilde{\mathbf{u}}_i + \gamma_1 \frac{\partial E}{\partial \tilde{\mathbf{u}}}(\tilde{\mathbf{u}}_i) + \gamma_2 \frac{\partial B}{\partial \tilde{\mathbf{u}}}(\tilde{\mathbf{u}}_i)$ $E = \tilde{\mathbf{e}}^T \tilde{\mathbf{e}}$ and B is the barrier function related to the constraints, and γ_1, γ_2 are proper designed step sizes.

if $Q(\cdot)$ is unknown $\tilde{\mathbf{u}}_{i+1} = \tilde{\mathbf{u}}_i + \gamma_1 \frac{\widehat{\partial E}}{\partial \tilde{\mathbf{u}}}(\tilde{\mathbf{u}}_i) + \gamma_2 \frac{\widehat{\partial B}}{\partial \tilde{\mathbf{u}}}(\tilde{\mathbf{u}}_i)$

- How to estimate the gradient with only input and output measurement
- When dimension of $\tilde{\mathbf{u}}$ is high: (1) High computation cost, (2) Complex estimator design

[3] Owens, D. H. (2016). Norm optimal iterative learning control. In Iterative Learning Control (pp. 233-276). Springer, London.

[4] Owens, D. H., Freeman, C. T., & Van Dinh, T. (2012). Norm-optimal iterative learning control with intermediate point weighting: theory, algorithms, and experimental evaluation. IEEE Transactions on Control Systems



Research Gaps

Current Limitation:

- If a directly calculation of gradient is impossible, how to design a **data-driven gradient descent ILC with input/output constraints** in the iteration domain?
- When the dimension of optimization variables ($\tilde{u}$) is high, it leads to a high computation cost and increases the complexity of designing a dither-based gradient estimator. How to **reduce the computation cost and improve the convergence speed?**



Aim 1 Data-driven ILC Framework

Problem: Data-driven gradient descent ILC for the constrained tracking problem

Current gap: The current NOILC requires the nominal model information

Approach:

- Barrier functions
- Dither-based gradient estimator
- Sufficient small design parameter ε
- Averaging technique

$$J(\tilde{\mathbf{u}}) = \tilde{\mathbf{e}}^T \tilde{\mathbf{e}} + B_1(\tilde{\mathbf{u}}) + B_2(\tilde{\mathbf{y}})$$



Aim 1 Data-driven ILC Framework

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$$\tilde{\mathbf{u}}_{i+1} = \tilde{\mathbf{u}}_i + \left[\sqrt{\Gamma\varepsilon} \cos(J(\tilde{\mathbf{u}}_i)) \mathbf{d}_i + \sqrt{\Gamma\varepsilon} \sin(J(\tilde{\mathbf{u}}_i)) \mathbf{q}_i \right]$$

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- Sufficient small design parameter ε
- Averaging technique

$$\tilde{\mathbf{u}}_{i+1} = \tilde{\mathbf{u}}_i + \sqrt{\Gamma}\varepsilon\cos(J(\tilde{\mathbf{u}}_i))\mathbf{d}_i + \sqrt{\Gamma}\varepsilon\sin(J(\tilde{\mathbf{u}}_i))\mathbf{q}_i$$

With $\varepsilon \ll 1$,

Input Updating $\tilde{\mathbf{u}}_i$ (Fast) Dithers $\mathbf{d}_i, \mathbf{q}_i$ (Slow)



Iteration scale separation



Aim 1 Data-driven ILC Framework

Problem: Data-driven gradient descent ILC for the constrained tracking problem

Current gap: The current NOILC requires the nominal model information

Approach:

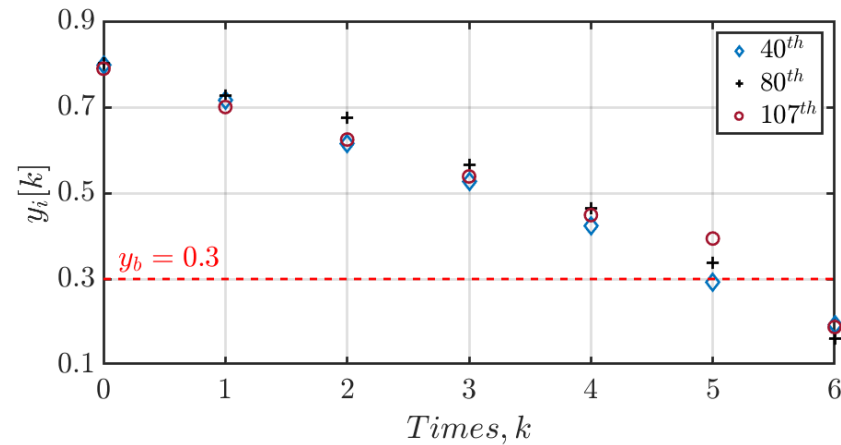
- Barrier functions
- Dither-based gradient estimator
- Sufficient small design parameter ε
- Averaging technique

$$\tilde{\mathbf{u}}_{i+1} = \tilde{\mathbf{u}}_i + \sqrt{\Gamma}\varepsilon\cos(J(\tilde{\mathbf{u}}_i))\mathbf{d}_i + \sqrt{\Gamma}\varepsilon\sin(J(\tilde{\mathbf{u}}_i))\mathbf{q}_i$$

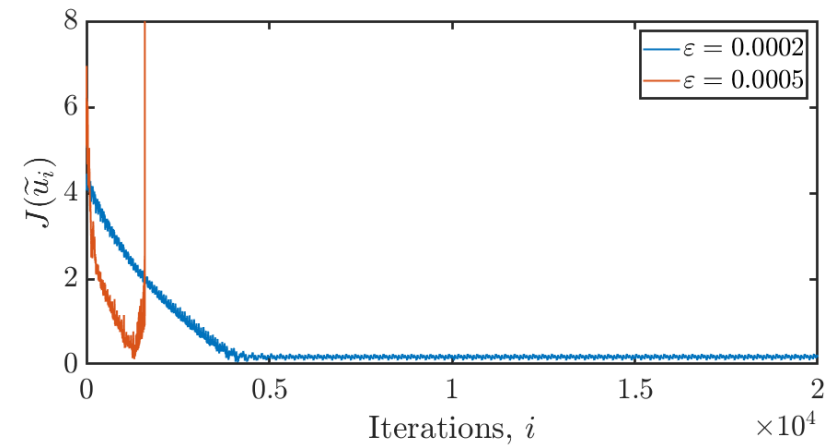
With sufficient small ε

Simulation Results

Simulation results from the published paper[5]



Output with unconstrained
data-driven gradient descent ILC



Cost with constrained data-
driven gradient descent ILC

- Constraints are satisfied due to the finite cost
- Tracking performance is guaranteed with sufficient small ε



Aim 2 Reduce the computation cost and improve the convergence speed

Current research gaps:

- High dimensional gradient calculation (Computation Cost)
- Complex dithers design when gradient estimator applied (Complexity)

Approach:

- (Special case) Point-to-point ILC framework^[6]

Input dimension decrease: All time instances → Time of interest

- (General case) ILC with Fast extremum seeking^[7]

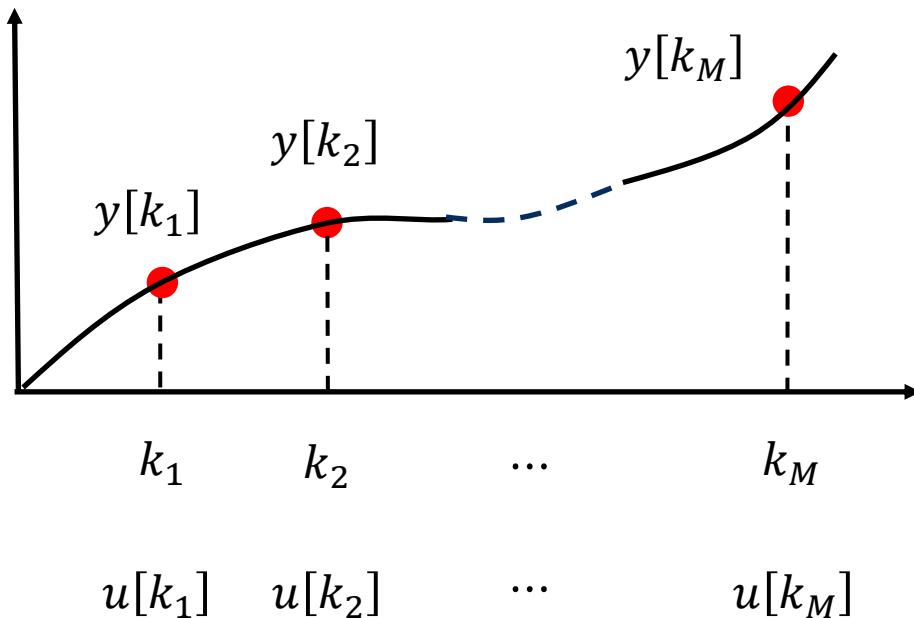
Faster convergence speed, same computation cost

[6] Freeman, C. T., & Tan, Y. (2012). Iterative learning control with mixed constraints for point-to-point tracking. IEEE Transactions on Control Systems Technology, 21(3), 604-616.

[7] Shekhar, R. C., Moase, W. H., & Manzie, C. (2014). Discrete-time extremum-seeking for Wiener–Hammerstein plants. Automatica, 50(12), 2998-3008.

Aim 2.1 Data-driven point-to-point ILC framework

Point to point setting:



Let $\mathbf{e}^\psi[l] = \mathbf{y}_r[k_l] - \mathbf{y}[k_l]$

- Time instances decrease:

$$N \rightarrow M, (N > M)$$

- New Cost defined:

$$J(\tilde{\mathbf{u}}) = \tilde{\mathbf{e}}^\psi{}^T \tilde{\mathbf{e}}^\psi \Rightarrow J[k_l] = \mathbf{e}^\psi[l]^T \mathbf{e}^\psi[l]$$

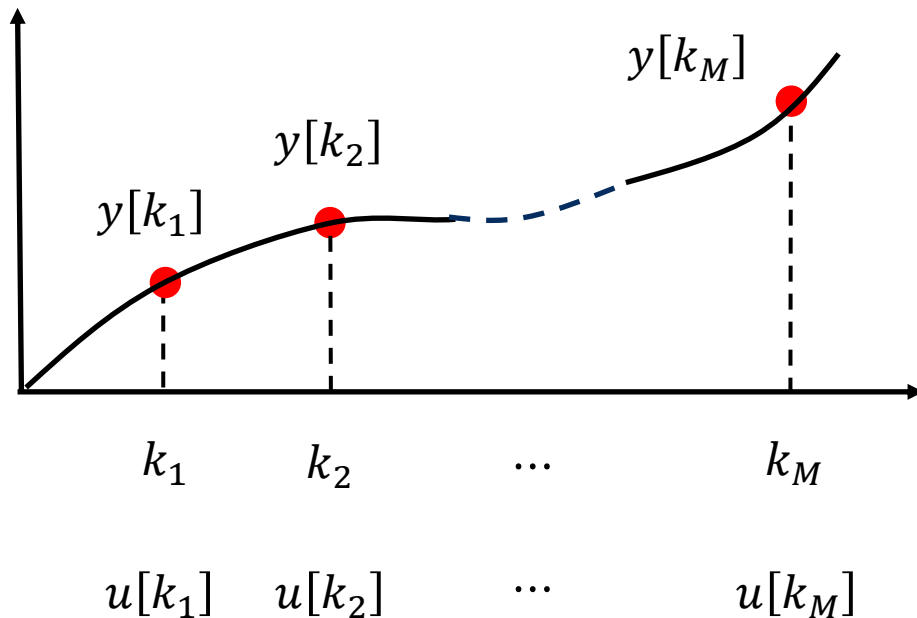
Needed Dither Sets: $p \cdot N$

J

$$[\mathbf{u}[0], \dots, \mathbf{u}[k_1], \dots, \mathbf{u}[k_M], \dots, \mathbf{u}[N-1]]$$

Aim 2.1 Data-driven point-to-point ILC framework

Point to point setting:



Let $\mathbf{e}^\psi[l] = \mathbf{y}_r[k_l] - \mathbf{y}[k_l]$

- Time instances decrease:

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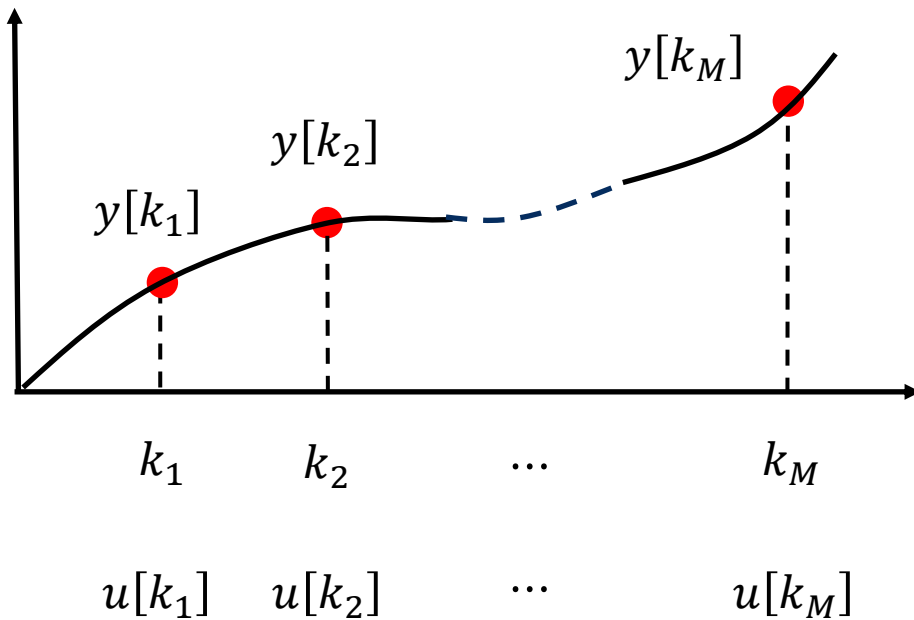
Needed Dither Sets: $p \cdot M$

J

$$[\mathbf{u}[0], \dots, \mathbf{u}[k_1], \dots, \mathbf{u}[k_M], \dots, \mathbf{u}[N-1]]$$

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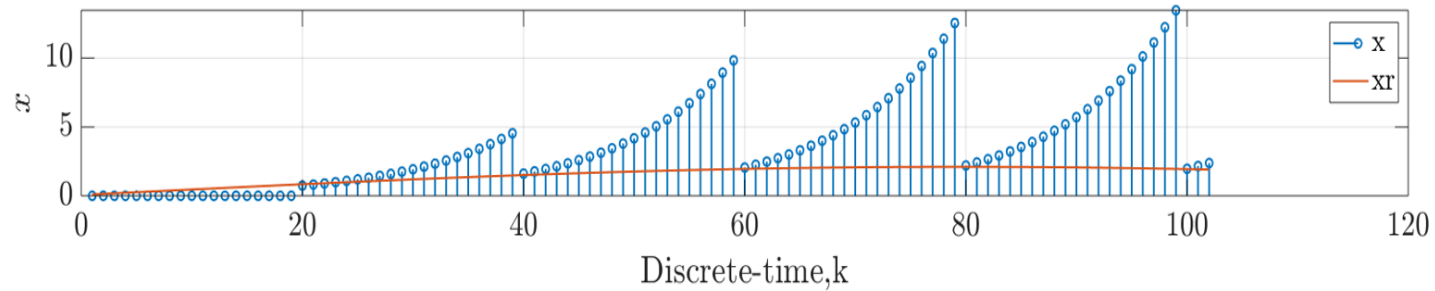
Needed Dither Sets: p

$$J[k_1] \quad \cdots \quad J[k_M]$$

$$[\mathbf{u}[0], \dots, \mathbf{u}[k_1], \dots, \mathbf{u}[k_M], \dots, \mathbf{u}[N-1]]$$

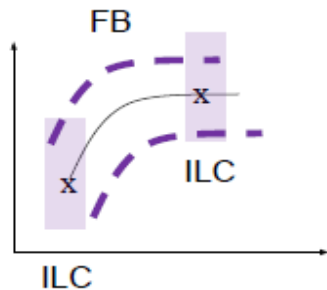
Aim 2.1 Data-driven point-to-point ILC framework

Coming questions: Outside the domain of attraction and sequential convergence



$$J[l] = \mathbf{e}^\psi[l]^T \mathbf{e}^\psi[l] = f(\mathbf{u}[k_l], \mathbf{x}[k_l])$$

- Control strategy: **Feedback** + Feedforward Control



Feedback: Bound the trajectory (Whole time interval)

Data-driven ILC: High precision tracking (Points interested)

Experiment Validation

